

Fall 2025 EE582 Homework 05 Report

Synchronous Machine Modeling and Transient Analysis

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Question 1

Problem Statement: Develop a qd0 model (in per-unit) of a 325-MVA Hydro-Turbine Generator (book p. 220) in Simulink using basic blocks. The generator is to be connected to an ideal source (infinite bus) that supplies balanced three-phase sinusoidal voltages at rated 1 pu. Assume all variables are in pu, unless otherwise specified. Assume positive direction of T_m is toward increasing the rotor speed ω_r and angle θ_r . Choose an ode solver with the settings (error tolerances and step-size limits) that you think produces trustable results and is numerically efficient. Comment on your choice.

Table 1: Hydro Turbine Generator Parameters

Parameter	Value
Rating	325 MVA
Line-to-line voltage	20 kV
Power factor	0.85
Poles	64
Speed	112.5 r/min
Combined inertia of generator and turbine	$J = 35.1 \times 10^6 \text{ J}\cdot\text{s}^2$, or $WR^2 = 833.1 \times 10^6 \text{ lbm}\cdot\text{ft}^2$
Inertia constant	$H = 7.5 \text{ s}$
Parameters in ohms and per unit:	
r_s	0.00234Ω , 0.0019 pu
X_{ls}	0.1478Ω , 0.120 pu
X_q	0.5911Ω , 0.480 pu
X_d	1.0467Ω , 0.850 pu
r'_{fd}	0.00050Ω , 0.00041 pu
X'_{fd}	0.2523Ω , 0.2049 pu
r'_{kq2}	0.01675Ω , 0.0136 pu
r'_{kd}	0.01736Ω , 0.0141 pu
X'_{lkq2}	0.1267Ω , 0.1029 pu
X'_{lkd}	0.1970Ω , 0.160 pu

Solution:

Simulink Model Development

ODE Solver Selection

Question 2

Problem Statement: Assume zero initial conditions and $E_{afd} = 0$. Implement a start-up transient similar to an induction machine (i.e., starting this synchronous machine from stall). Plot variables i_{as} , T_e , ω_r . Comment on the results. Explain how it would be practical to bring a large machine like this online.

Solution:

Simulation Results

Analysis of Start-up Transient

Practical Start-up Procedure

Question 3

Problem Statement: Calculate the initial conditions that would correspond to an idle steady-state operation with $T_m = 0$ and excitation $E_{x_{fd}} = 1.0$ pu, and set the initial conditions appropriately in your model. Implement the following transient study:

- At $t = 1$ s, the excitation $E_{x_{fd}}$ is stepped down to 0.8 pu.
- Then, at $t = 7$ s, the excitation $E_{x_{fd}}$ is stepped up to 1.2 pu.
- Then, at $t = 15$ s, the mechanical torque is stepped to $T_m = 0.85$ pu supplying power from the mechanical system to the electrical system, and the model is continued to run till $t = 25$ s.

Plot variables i_{as} , i_{fd} , i_{kq} , T_e , ω_r , θ_r (in electrical degrees), as well as real power P_e and reactive power Q_e . Based on the transients observed, briefly explain the effect of the excitation and torque on the active and reactive power and the type of transients seen.

Solution:

Initial Conditions Calculation

Simulation Results

Analysis of Excitation Effects

Analysis of Torque Effects

Question 4

Problem Statement: Now assume that the q-axis damper winding is damaged. This can be implemented by setting the resistance of this winding to a very high value, e.g., set $r_{kq} = 1$ pu. Repeat what you did in part 3).

Solution:

Model Modification

Simulation Results

Comparison with Question 3

Question 5

Problem Statement: Now assume that the d-axis damper winding is damaged. Repeat what you did in part 3). No need to plot anything. Just make observations.

Solution:

Model Modification

Observations

Key Differences from Question 3

Question 6

Problem Statement: Based on the observations in parts 4) and 5), explain the effect and the role of each of these damper windings.

Solution:

Role of the q-axis Damper Winding

Role of the d-axis Damper Winding

Overall Function of Damper Windings
